MLP Project Milestone 3: Feature Engineering and Feature Selection

To understand about all the milestones in detail, Please *click here*

Roll number	

Submission guidelines

1. **Make your own copy with File > Make a copy, and write only in your copy.**
2. **Once you have finished writing your report, save or export the .docx document as a PDF.**
3. **Rename the PDF file using the following exact convention: [roll-number]_26t3_M[Milestone-Number].pdf**
4. **Example: If your roll number is 21f1001234 and you are submitting the PDF for Milestone 3, your file must be named exactly: 21f1001234_26t3_M3.pdf**

Instructions

- For all the questions present in the document, please use the dataset which is provided on kaggle for the current term mlp project.
- Write each answer in the "Your answer" box under the question, and paste the Python code (wherever required) you ran into the "Your code" box as text, not as a screenshot.
- Support every claim with evidence from your own analysis of the competition data: a number, a small table or a plot (Insert > Image). Summarise long outputs instead of pasting them in full.
- The points under each question describe what an answer should cover. If a point does not apply to your dataset (for example, there are no date columns), you can skip it.
- Write in your own words. Clear reasoning backed by evidence matters more than length of your answer.
- There is no single correct answer, these questions are subjective and there can be multiple valid explanations of questions. Use your interpretation and understanding to answer the questions.
- You need not include all the columns(features) while answering questions, you can select a few meaningful columns and do analysis on them. Try to cover at least 50% of the columns(features) for the questions in this milestone.
- You MAY use a different kaggle notebook instead of the one shared with the host to do the task given in milestones.
- You do not need to analyse all the columns;however, make sure to cover a good number of columns to support your analysis and decisions. Complete feature engineering and feature selection need not be covered during this milestone.
- The answer length can vary depending on questions. Keep your answer concise and give a summary of the analysis and cover details using visualizations/bullet points wherever required. The maximum length of answer can be considered around 100 to 400 words.
- **Plots, graphs and figures:** add them in the "Figures" section at the end of this document, under the heading of the question they belong to, not inside the answer boxes. Number each figure as Figure Q<question number>.<figure number> (for example, Figure Q3.1) and refer to it by this number in your answer (for example, "see Figure Q3.1").

Q1. Feature engineering plan

Explain the role of feature engineering and feature selection in machine learning. Based on the EDA findings, identify opportunities to create new features that may be useful for model building.

One suggested approach would be as follows:

- Explain the difference between preprocessing, feature engineering, and feature selection.
- Identify limitations in the existing features based on the EDA findings.

Write your answer here.

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# Paste the code you ran for this question
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Q2. Features from existing columns

Create new features by combining existing columns to capture additional information that may be useful for model building. Explain the reasoning behind the features created.

One suggested approach would be as follows:

- Explore feature combinations such as ratios, differences, sums, rates, or other meaningful calculations.
- Explain the reason for selecting the combinations based on EDA findings or domain knowledge.
- Evaluate whether the new features improve the model's predictive performance.

Write your answer here.

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# Paste the code you ran for this question
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Q3. Capturing non-linear relationships

Investigate whether any numerical features have non-linear relationships with the target. Apply suitable feature engineering techniques to capture these relationships and evaluate their usefulness.

One suggested approach would be as follows:

- Identify possible non-linear relationships using EDA, visualisations, or other appropriate methods.
- Compare the methods and evaluate their effect on model performance.

Write your answer here.

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# Paste the code you ran for this question
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Q4. Capturing feature interactions

Investigate whether the effect of one feature on the target depends on another feature. Create suitable interaction features and evaluate whether they improve the model's predictive performance.

One suggested approach would be as follows:

- Identify potential interactions between features using EDA or domain knowledge.
- Create suitable interaction features, such as products of two features or other meaningful combinations.
- Compare model performance with and without the interaction features.

Write your answer here.

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# Paste the code you ran for this question
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Q5. Feature extraction from other complex data types

Create additional features by extracting useful information from other complex data types, such as dates, text, lists, geographical information, wherever applicable to the dataset.

One suggested approach would be as follows:

- Identify the columns that contain useful information not fully represented by the existing features.
- Apply suitable feature extraction techniques, such as date-based calculations, text-derived features, list statistics, or geographical features.
- Explain how the extracted features may help predict the target and evaluate their effect on model performance.

Write your answer here.

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# Paste the code you ran for this question
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Q6. Group-based feature engineering

Investigate whether grouping related rows can provide useful information for predicting the target. Create suitable group-based features and explain how data leakage is prevented.

One suggested approach would be as follows:

- Identify meaningful groups of rows based on features such as location, category, or another relevant attribute.
- Create aggregate features using statistics such as counts, means, medians, or standard deviations.
- Explain methods used to calculate these features without using information from validation or test targets or introducing other data leakage.
- Evaluate whether the group-based features improve model performance.

Write your answer here.

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# Paste the code you ran for this question
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Q7. Identifying and handling redundant features

Identify redundant or strongly related features in the dataset, including newly engineered features. Explain the features which are removed, combined, or retained and evaluate the effect of these decisions.

One suggested approach would be as follows:

- Identify potentially redundant features using correlation analysis or other suitable methods.
- Explain reasoning behind features being removed, combined, or retained.
- Compare model performance and the number of features before and after handling redundancy.
- Discuss how redundant features may affect different types of models.

Write your answer here.

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# Paste the code you ran for this question
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Q8. Comparison of feature selection methods

Apply and compare suitable feature selection methods to identify the features which are useful for predicting the target. Explain the selection of the final set of features and provide justification for it.

One suggested approach would be as follows:

- Apply suitable feature selection methods, such as filter methods, wrapper methods.
- Explain the decision behind choosing the certain number of features and justify the final choice based on model and performance on validation set.
- Explain the handling of train-test leakage when applying feature selection techniques.

Write your answer here.

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# Paste the code you ran for this question
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Q9. Dimensionality reduction

Investigate whether dimensionality reduction can provide a more compact representation of features. Compare this approach with selecting a subset of the original features and determine which is more suitable for the dataset.

One suggested approach would be as follows:

- Explain the difference between feature selection and dimensionality reduction.
- Explore a suitable dimensionality reduction technique, such as Principal Component Analysis (PCA), and explain the reasoning behind choosing a certain number of components.
- Compare model performance using the original features, selected features using feature selection methods, and reduced features using dimensionality reduction.

Write your answer here.

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# Paste the code you ran for this question
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Q10. Final feature set and model performance

Summarise final feature engineering decisions. Evaluate their overall effect by comparing with the model trained before and after the final feature set.

One suggested approach would be as follows:

- Summarise the features created, removed, and retained, along with the final number of features.
- Compare the model's performance before and after feature engineering and selection using the same validation strategy and evaluation metric.
- Identify which feature engineering or selection techniques contributed most to the final model's performance.

Write your answer here.

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# Paste the code you ran for this question
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Figures

Add all plots, graphs and figures for this milestone in this section, under the heading of the question they belong to.

How to add a figure

1. Place your cursor below the heading of the question the figure belongs to, and use Insert > Image to add the figure.
2. Number each figure in the format Figure Q<question number>.<figure number>. For example, the first two figures for Q3 are Figure Q3.1 and Figure Q3.2.
3. Write the figure number and a short title on the line just above the figure, for example: Figure Q3.1: Relationship between a numerical feature and the target.
4. In your answer, refer to the figure by its number, for example: "The relationship is not linear across the range of the feature (see Figure Q3.1)."
5. Resize each figure so that it fits within the page width and all axis labels and legends are readable.
6. Replace the placeholder lines under each heading with your figures. If a question has no figures, leave its heading as it is.

Figures for Q1. Feature engineering plan

Figure Q1.1: [Short title of the figure]

[Insert image here]

Figures for Q2. Features from existing columns

Figure Q2.1: [Short title of the figure]

[Insert image here]

Figures for Q3. Capturing non-linear relationships

Figure Q3.1: [Short title of the figure]

[Insert image here]

Figures for Q4. Capturing feature interactions

Figure Q4.1: [Short title of the figure]

[Insert image here]

Figures for Q5. Feature extraction from complex data types

Figure Q5.1: [Short title of the figure]

[Insert image here]

Figures for Q6. Group-based feature engineering

Figure Q6.1: [Short title of the figure]

[Insert image here]

Figures for Q7. Identifying and handling redundant features

Figure Q7.1: [Short title of the figure]

[Insert image here]

Figures for Q8. Comparison of feature selection methods

Figure Q8.1: [Short title of the figure]

[Insert image here]

Figures for Q9. Dimensionality reduction

Figure Q9.1: [Short title of the figure]

[Insert image here]

Figures for Q10. Final feature set and model performance

Figure Q10.1: [Short title of the figure]

[Insert image here]